Systematic Options Volatility Modeling: Implied vs Realized Volatility in the Indian Options Market

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***Abstract*—This project builds a Black-Scholes-based options analytics and research pipeline: correct implementations of European option pricing and all five standard Greeks, a numerical implied-volatility solver, realized-volatility estimation, and a systematic comparison of implied vs. subsequently realized volatility. Long and short at-the-money straddle strategies are backtested with an honest treatment of the fundamentally asymmetric risk of short-volatility positions. Since complete and clean historical Indian options-chain data was not freely available in a convenient, research-ready format, results are demonstrated using a documented synthetic option-chain history. This synthetic dataset utilizes a geometric Brownian motion underlying, known true volatility, mean-reverting implied volatility, realistic documented skew, and Black-Scholes repricing. These results are intended for controlled methodological validation and not as evidence of real-market profitability. The pipeline is explicitly built with three separated modes (research/ demo, historical CSV, live) so it can be re-run on real market data without any code changes to the statistical methodology.**

I. Introduction

Options give traders exposure to volatility itself, not just the direction of the underlying. A central question in derivatives trading is whether the market's own volatility forecast — implied volatility, extracted from option prices via the Black-Scholes model — is a good predictor of the volatility that subsequently actually occurs (realized volatility). If implied volatility systematically overstates realized volatility, sellers of options (who are effectively short volatility) earn a persistent "volatility risk premium" on average, at the cost of occasional large losses when volatility does spike. This project builds the tools to investigate that question directly and to backtest simple strategies that attempt to harvest (or bet against) that premium.

II. Background

The Black-Scholes-Merton model (Black & Scholes, 1973; Merton, 1973) gives a closed-form price for a European option as a function of five inputs: spot price, strike, time to expiry, risk-free rate, and volatility. Because the first four are directly observable, market option prices can be inverted to solve for the volatility the market is implicitly pricing in — the implied volatility. Comparing this to realized (historical, backward-looking, or subsequently-realized, forward-looking) volatility is a standard practitioner and academic exercise in volatility trading (Sinclair, 2013).

III. Problem Statement

Given historical (or synthetic, when historical is unavailable) underlying prices and option chain data, (a) correctly price options and their Greeks, (b) extract implied volatility from option prices, (c) compare implied volatility to realized volatility, and (d) determine whether a systematic straddle strategy built on this comparison is profitable under the stated assumptions — while being explicit about the asymmetric risk that short-volatility strategies carry.

IV. Research Question

Does the spread between an option's implied volatility and the volatility subsequently realized by the underlying carry useful information for a systematic options trading strategy — and how does that translate into the asymmetric risk of long vs. short volatility positions?

V. Objectives

1. Estimate realized volatility from underlying prices.
2. Implement Black-Scholes pricing and all five Greeks correctly.
3. Build a robust numerical implied-volatility solver.
4. Compare implied volatility against subsequently realized volatility.
5. Visualize IV skew (across strikes) and term structure (across expiries).
6. Backtest long and short ATM straddle strategies using a simplified cost model.
7. Report full, honest performance metrics, including the asymmetric tail risk of short volatility strategies.
8. Explicitly separate and document research/demo, historical-backtest, and live-data modes.

VI. Literature / Conceptual Background

- **Black & Scholes (1973) / Merton (1973)**: the foundational option-pricing model used throughout this project.
- **Volatility risk premium**: a widely documented stylized fact in equity index options markets, where average implied volatility tends to exceed average subsequently realized volatility, compensating option sellers for bearing tail (jump/crash) risk.

VII. Dataset

Three modes, matching the project's data-availability requirements:

1. **Research/demo (synthetic)**: Synthetic data was selected because complete historical Indian options-chain data with strike-wise, expiry-wise, and CE/PE information was not freely available in a consistently clean and research-ready format. This dataset features a GBM underlying, known annualized volatility, mean-reverting ATM IV, documented IV skew, and Black-Scholes repricing.
2. **Historical CSV**: user-supplied historical option-chain data.
3. **Live (yfinance)**: real underlying price history for any yfinance-supported ticker, and separately, a current-day options snapshot.

VIII. Data Collection & Preprocessing

Implemented in `src/data_collection.py`, exposing all three modes as separate functions. For preprocessing, underlying prices drop any missing observations. Option chains drop rows missing price, strike, or expiry information rather than imputing option prices, since a wrong option price directly corrupts the IV solver's output.

IX. Methodology

The research methodology is separated from the data source, meaning historical real-market option-chain data can be supplied through the Historical CSV mode without changing the core Black-Scholes pricing, Greeks, implied-volatility solver, realized-volatility estimation, IV-RV comparison, or strategy backtesting methodology.

1. **Realized volatility**: rolling standard deviation of log returns over a window, annualized by $\sqrt{252}$.
2. **Black-Scholes pricing and Greeks**: implemented directly from the closed-form formulas, vectorized over strikes/expiries.
3. **Implied volatility solver**: Brent's method on the pricing function, solving for the σ that reprices Black-Scholes to the observed market price.
4. **IV vs. RV comparison**: at each date, compare the current ATM implied volatility against the volatility subsequently realized by the underlying.
5. **Skew and term structure**: snapshot the IV across strikes (skew) and across expiries (term structure).
6. **Strategy backtest**: enter an ATM straddle on a schedule of entry dates, hold for a fixed number of days, then mark to market.

IV vs RV Comparison

Fig. 1. Time series comparison of Implied Volatility (IV) and subsequently Realized Volatility (RV).

X. Mathematical Formulation

Black-Scholes-Merton pricing:

```
d1 = [ln(S/K) + (r - q + sigma^2/2) * T] /
(sigma * sqrt(T))
d2 = d1 - sigma * sqrt(T)
Call = S * e^(-qT) * N(d1) - K * e^(-rT) *
N(d2)
Put  = K * e^(-rT) * N(-d2) - S * e^(-qT) *
N(-d1)
```

Greeks:

```
Delta_call = e^(-qT) * N(d1)
Delta_put = -e^(-qT) * N(-d1)
Gamma = e^(-qT) * phi(d1) / (S*sigma*sqrt(T))
Vega = S * e^(-qT) * phi(d1) * sqrt(T)
```

Implied volatility: the root, in sigma, of `BS_price(S,K,T,r,sigma) - market_price = 0`, solved via Brent's method.

Realized volatility: `RV = std(log_returns) * sqrt(252)` over a rolling window.

Volatility Skew

Fig. 2. Implied Volatility Skew (Volatility Smile), demonstrating higher implied volatility priced into out-of-the-money puts.

XI. System Architecture & Implementation

See `docs/architecture.md`. In brief: `data_collection.py` → `black_scholes.py` + `volatility.py` → `strategy_backtest.py`. Implemented in modular Python (`src/`) with 18 passing unit tests used specifically to validate numerical correctness. These tests cover Black-Scholes pricing, Greeks (checked against both analytical identities like put-call parity and numerical finite-difference derivatives), and the IV solver.

XII. Experimental Setup

- Synthetic underlying: 400 trading days, GBM with true annualized volatility 24%, drift 8%.
- Synthetic option chain: 7 strikes (90%–110% moneyness) × 5 expiries × 2 option types, generated daily.

- Target strategy expiry: 30 days; holding period: 14 days; new entry every 20 trading days.
- Transaction cost: For the synthetic-data backtest, transaction costs were modelled as a simplified 2% of option premium on both entry and exit. This is a modelling assumption rather than a claim about actual Indian-market transaction costs.
- Risk-free rate: 7% (approximate short-term Indian G-Sec yield).

XIII. Results

- Realized volatility (20-day window) averaged ≈22.7%, close to the known true GBM volatility of 24.0%.
- ATM 30-day implied volatility averaged ≈23.9% over the sample, ranging 17.1%–30.8%.
- Mean(IV − subsequent realized vol) ≈ +0.09 percentage points. The reported mean IV−RV spread is computed by first calculating the date-level difference between current ATM implied volatility and subsequently realized volatility and then averaging these differences across observations. Therefore, it is not necessarily equal to the difference between the separately reported sample averages of IV and RV.
- Strategy backtest: The synthetic-data experiment produced the following results: the long straddle had a 38.9% win rate, average P&L per trade ≈ −₹2.80, and profit factor ≈ 0.68.
- The short straddle had a 61.1% win rate, average P&L per trade ≈ +₹2.80, but a worst-trade loss of ≈ −₹50.8 against a best-trade gain of only ≈ +₹23.4. These results should not be interpreted as evidence of live-market profitability in the real Indian options market, but rather as a demonstration of the asymmetric risk profile of long versus short volatility strategies.

P&L Distribution

Fig. 3. Short Straddle Per-Trade P&L Distribution. Highlights the asymmetric risk profile of short-volatility strategies.

XIV. Performance Evaluation & Discussion

The short straddle's higher win rate and positive average P&L look attractive in isolation, but the per-trade P&L distribution (Section XIII, Fig. 3) shows its worst trade lost more than double its best trade gained. Reporting only the average, win rate, or profit factor without this distribution would be materially misleading.

XV. Risk Analysis

- **Unbounded short-option risk**: In a short straddle, the premium received allows for frequent smaller gains; however, a large underlying move can produce disproportionately large losses. This demonstrates that a high win rate does not automatically mean good risk-adjusted performance, and average P&L alone can hide tail risk. Analyzing the worst-trade/best-trade distribution is critically important.
- **Margin risk**: exchanges require substantial margin for uncovered short option positions in practice; this project does not model margin requirements.
- **Vega/Gamma risk**: a straddle's value is highly sensitive to changes in implied volatility (vega) and accelerates near expiry (gamma).

XVI. Limitations

Synthetic-Data Baseline: The reported experimental results are based on synthetic option-chain data. Therefore, the numerical strategy performance should be interpreted as a controlled methodology demonstration rather than evidence of real-world profitability. Synthetic data cannot fully reproduce actual market microstructure, discrete strike/expiry listings, bid-ask spreads, liquidity constraints, slippage, event-driven volatility jumps, or actual margin requirements.

Modelling Constraints: Furthermore, the backtest uses a simplified flat-percentage transaction cost model rather than true bid-ask-spread-based costs, no margin modeling, and a single flat risk-free rate rather than a real yield curve.

XVII. Future Work

Realistic extensions to this research pipeline include: applying the methodology to reliable historical NIFTY/BANKNIFTY option-chain data; substituting the flat transaction-cost assumption with actual bid-ask spreads and slippage; modelling exchange margin requirements; incorporating a time-varying Indian risk-free yield curve instead of a flat rate; including event-driven volatility and jumps; and testing additional volatility strategies over longer out-of-sample periods.

XVIII. Conclusion

This project successfully demonstrates a rigorous options analytics pipeline, including Black-Scholes pricing, Greeks calculation, an IV solver, realized-volatility estimation, IV vs. subsequent RV analysis, a straddle backtesting framework, and asymmetric short-volatility risk analysis. It does NOT establish profitability in the actual Indian options market, a statistically proven Indian volatility risk premium, or live trading profitability. Applying this validated pipeline to reliable historical Indian option-chain data represents the natural next step for future research.

References

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3. Sinclair, E. (2013). *Volatility Trading.* Wiley.